Who Backs Into the Bay? Benchmarking Negotiation Policies for Head-On Robot Encounters in Single-Lane Corridors

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***Abstract*—When two autonomous mobile robots meet head-on in a corridor that admits only one, one of them must reverse into a pull-in bay or out of the corridor. We study who should yield, and into which refuge, with a benchmark that runs the same agent code in a 2D unicycle simulator and in ROS 2/Gazebo with TurtleBot3 Waffle robots. We compare a non-communicating reactive baseline, fixed priority, an ordered rulebook, first-come corridor reservation, cost bidding, a group bid for convoys, a Clarke-pivot mechanism, and a learned tabular negotiator against the best forced yielder, a hindsight reference that tries both yield decisions. In 2 940 single-bay episodes the reactive baseline succeeds in 93.8% and incurs 3.6× the reference's total delay; the negotiating policies succeed in every episode. Cost bidding with exact costs matches the reference; with unequal urgency its urgency-weighted excess delay is 0.04 s against 27.1 s for fixed priority, and with log-normal bid noise of $\sigma = 0.5$ it is no better than a noise-free closest-refuge rule. Lower-bid-yields bidding can be gamed; with Clarke-pivot payments no unilateral misreport we tested pays off under the predicted costs, at the price of a budget surplus and no protection against collusion. Reservation is slower than fixed priority in our layout. Under 90–99% packet loss, a factorial ablation of three protocol safeguards (5 760 episodes) observed no collisions in any arm. We extend the benchmark to convoys, a T-junction, inconsistent and wrong maps, and 158 Gazebo runs, which agreed with 2D on the first yielder once two sim-to-real gaps were fixed. Code and per-episode results are released.**

***Index Terms*—Multi-robot systems, negotiation, deadlock, path planning for multiple mobile robots, benchmarking.**

I. Introduction

NARROW aisles, doorways, and service corridors force autonomous mobile robots (AMRs) into conflicts that local collision avoidance cannot resolve: if two robots meet head-on where only one fits, someone must back off. Reciprocal collision avoidance and control-barrier-function controllers deadlock in such symmetric configurations [1], [2], [3], while centralized planners and traffic managers avoid the conflict at the price of a global plan [4], [5], [6], [7], [8], [9]. Between these extremes lies decentralized negotiation: the robots exchange messages and agree who yields. Rules of the road [10], [11], auctions [12], [13], learned policies [14], [15], language-model negotiation [16], [17], and foundation-model high-level planners for deadlock resolution [18] have all been proposed, but each paper builds its own scene and metrics, and no study compares the families under the same physics and communication model [19].

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DRAFT, not for distribution.

This letter contributes:

1) an open benchmark of the head-on corridor encounter with pull-in bays, in which the same agent state machine runs in a 2D simulator for Monte Carlo sweeps and in ROS 2 Jazzy/Gazebo Harmonic with TurtleBot3 Waffles;
2) a comparison of reactive, fixed-priority, rule-based, reservation, cost-bidding, group-bidding, Clarke-pivot, and learned negotiators over bay geometry, robot asymmetry, bid noise, strategic misreporting, repeated encounters, and packet loss and latency, all on one code version with per-episode results;
3) a factorial ablation of the protocol safeguards under extreme packet loss, and extensions to convoys, a T-junction, inconsistent maps, and physically blocked bays;
4) a Gazebo check (158 runs) of where the 2D conclusions transfer, which exposed two sim-to-real gaps.

II. Related Work

Deadlock in narrow passages. Symmetric encounters are a known failure mode of decentralized avoidance. Grover et al. [1] characterize CBF deadlocks; Dergachev and Yakovlev [2] switch to local multi-agent path finding (MAPF) when ORCA stalls; Chandra et al. [3] perturb controls onto liveness sets in doorways and hallways. These works resolve who goes first by slowing down, not by choosing which robot reverses into which refuge. *Centralized coordination.* MAPF and AGV traffic management treat bays as graph nodes [20], [5], [21], [22]; safe-interval planning reserves corridor segments in time [6], fleet managers such as Open-RMF negotiate conflicting schedules centrally [9], [8], and precedence-preserving coordinators provide safety under delays [4], [23]. They require a global planner or schedule database and typically charge reversing as ordinary motion. *Negotiation and mechanisms.* Social laws [11], market-based allocation [12], and negotiated path planning [13] assume truthful costs. Vickrey–Clarke–Groves mechanisms [24], [25], [26], [27] make truthful reporting dominant, but no efficient, truthful mechanism is budget balanced in general [28], [29]. Karma schemes trade non-monetary credit across repeated encounters to make right-of-way allocation fair [30], [31]; cooperative give-way at road bottlenecks has also been studied for connected vehicles [32]. *Learning and LLMs.* MARL give-way tasks exist in VMAS [14]; MARLIN uses LLM negotiation to guide MARL training in narrow-corridor swaps, with

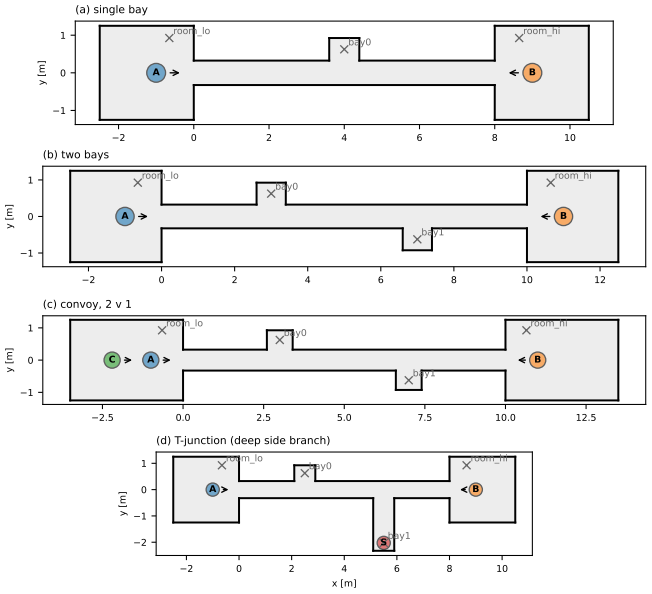


Fig. 1. Benchmark scenarios (gray: free space; ×: refuge parking poses). Arrows show travel direction; in (d) robot S starts in the side branch and A's destination is the branch.

physical TurtleBot3 experiments [16]; GameChat sets priority through LLM dialogue in doorways [17]. None compares against rule-based, bidding, and hindsight baselines under communication impairment, which is the gap we address.

III. Benchmark

A. Scenarios

The base scenario (Fig. 1a) is an 8 m×0.65 m corridor between two 2.5 m rooms. A TurtleBot3 Waffle (0.25 m swept radius, 0.26 m/s forward, 0.15 m/s reverse) fits; two cannot pass. Bays are 0.8 m wide and 0.6 m deep, with configurable position, side, and number; the end rooms also serve as refuges. Robot A enters at $t = 0$, and B's start delay is drawn from $U[0, 8]$ s. Variants add a second bay (Fig. 1b), a two-robot convoy against one oncoming robot (Fig. 1c), and a T-junction (Fig. 1d) whose side branch is approximated as a 2 m-deep bay that a robot may leave or have as its destination; it is not a full 2D junction. Robots can differ in urgency, payload, and speed limits. Communication is a broadcast bus with configurable latency, jitter, and loss.

B. Agent and protocol

Each robot runs the same state machine (cruise, negotiate, yield, proceed, done) at 20 Hz, with lidar perception and a unicycle model. A conflict is detected when an oncoming peer is within the detection range (6 m by default). Robots broadcast STATE messages at 5 Hz and, on conflict, exchange BID messages containing a snapshot of their pose, speed, urgency, payload, and known refuges. Both robots then run the same deterministic decision function on the same pair of snapshots, so they agree without a third message when no message is lost. A yield plan names the yielder and its refuge; feasibility checks that the refuge lies behind the yielder and is not taken.

With message loss the robots may decide on different snapshots. Three safeguards address this, and each can be switched off for ablation. *Commit*: each robot broadcasts its decision and acts only once it holds the peer's, the lower-ID robot's decision wins a disagreement, and an unconfirmed decision expires after a timeout. *State guard*: a yielder whose peer's STATE shows it yielding into the same refuge, or yielding to it, gives way and proceeds instead. *Bay check*: a bay that the lidar shows occupied counts as taken, and a yielder that finds a robot in its bay proceeds. A robot waiting to pass also renegotiates if its supposed yielder approaches it. We give no formal safety or liveness proof for this protocol; its behavior is characterized empirically in Sec. IV-C.

C. Policies

Reactive: no messages; a blocked robot waits a random patience time and then backs to the nearest refuge behind it. *Fixed priority*: the higher-ID robot always yields. *Rulebook*: an ordered list of rules (no feasible refuge; urgency; payload; first in corridor; refuge within 2 m behind; ID tie-break), the first discriminating rule decides; a variant puts "closest refuge yields" first. *Reservation*: first come, first served; a robot still outside the corridor while an oncoming robot is inside waits in its end room, and robots that are both inside fall back to the rulebook. `cost_bid`: for each candidate yielder Y with proceeder P, both robots predict the delays d_Y, d_P of Y's best feasible plan from the robot kinematics and evaluate $u_Y d_Y + u_P d_P$, where u is urgency; the cheaper outcome is chosen. `group_bid`: as `cost_bid`, but a side's cost includes each yielding member's own urgency-weighted delay, the wait of the yielding side while the other side passes, and the delay of the proceeding side's followers. *Clarke*: each robot reports its predicted urgency-weighted delay $r_i(o)$ under both outcomes o; the mechanism chooses $o^* = \arg\min_o \sum_i r_i(o)$ and charges robot i the Clarke pivot $p_i = \sum_{j \neq i} r_j(o^*) - \min_o \sum_{j \neq i} r_j(o)$ in delay credit [25]. *Learned*: a tabular contextual bandit over a coarse observation (each robot's binned distance to its nearest feasible refuge behind it, which robot is more urgent, and both payload flags), trained on 800 encounters by running both forced outcomes; unseen observations fall back to fixed priority. It is a lightweight learned baseline, not deep multi-agent RL.

D. Metrics and statistics

We report success (all robots reach their goals before the time limit without collision), deadlock (time limit reached), collision (overlap of the Waffle footprint rectangles), total delay relative to free-flow travel time, urgency-weighted delay, reverse distance, and Jain's fairness index [33].

Excess delay. The delay of an episode is $D = \sum_i w_i \, (T_i - T_i^{\text{free}})$, where T_i is robot i's travel time, T_i^{free} its travel time with the corridor to itself, and $w_i = 1$ (total delay) or w_i equal to the robot's urgency (weighted delay). For every two-robot encounter k we also run the two forced outcomes, A yields and B yields, each into that robot's best feasible refuge; the

better of the two is the *best forced yielder*. The excess delay of policy π over N encounters is

$$\Delta_\pi = \frac{1}{N}\sum_{k=1}^{N}\Big(D_\pi^{(k)} - \min\big(D_A^{(k)}, D_B^{(k)}\big)\Big), \tag{1}$$

paired by encounter, with the same weights w_i in all three terms. $\Delta_\pi = 0$ means the policy always made the better of the two yield decisions. The reference chooses only who yields, so Δ_π measures the quality of that decision, not the distance to a globally optimal joint motion plan, and a policy that also chooses *when* to move could in principle have $\Delta_\pi < 0$. It is the ex-post regret of decision theory; we avoid that term because it has different meanings in online learning and in mechanism design. Excess delay is defined for two-robot encounters; for the convoy we add a side-level reference, the better of "the convoy yields" and "the lone robot yields".

Statistics. All 2D results come from one code version, and every per-episode result is released with the scripts that produce the tables. A seed fixes the start delays and the communication noise; at 0% loss the negotiating policies are deterministic given the start delays, so the independent samples are the distinct draws, and policies are compared on the same draws. Delays are averaged over successful episodes only; failed episodes are censored at the time limit and reported through the success, deadlock, and collision rates. $\pm$ values are 95% confidence intervals of the mean; because Δ_π is a per-encounter paired difference, its interval is a paired interval. Rates carry Wilson 95% intervals, and an observed count of zero is reported with its 95% upper bound.

IV. Results

A. Two robots, one corridor

Price of no communication. Over 2 940 single-bay episodes (60 draws at each of 7 bay positions) the reactive baseline succeeded in 93.8% [91–96] of episodes; the rest deadlocked. Its total delay over successes was 97 ± 5 s against 26.8 ± 0.6 s for the best forced yielder, it reversed 5.7 m against 0.64 m, and both robots backed off in 39% of episodes. All negotiating policies succeeded in all 420 of their episodes without collision.

Price of decentralization and bay position. With symmetric robots, fixed priority and the default rulebook make identical decisions, with 6.4 ± 0.6 s excess delay, rising to 14.6–14.8 s when the bay is 2–3 m from A's entrance and the late robot B is sent back past it. Putting "closest refuge yields" first reduces the excess delay to 3.1 ± 0.5 s. `cost_bid` chose the reference's yielder in every single- and two-bay episode. This match is a best case, because the bids use the simulator's own kinematics.

Reservation. Over 210 encounters (bay at 1–7 m, 30 draws each), first-come reservation succeeded in every episode but had 18.8 ± 0.9 s excess delay, against 6.5 ± 0.9 s for fixed priority and 0 for `cost_bid`, and it beat the best forced yielder in none of the 210 encounters. In this layout B starts on A's exit path, so waiting for the corridor means pulling into its end room rather than stopping at the entrance; layouts with a holding point outside the exit path would favor reservation.

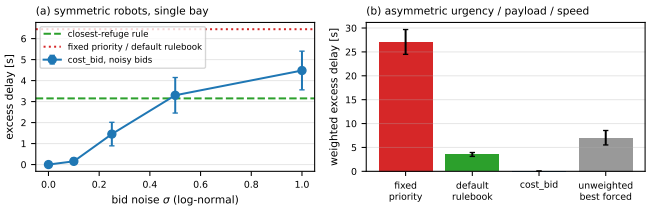


Fig. 2. (a) Excess delay (1) of `cost_bid` over the best forced yielder as bid noise grows, compared with two noise-free rulebooks. (b) Urgency-weighted excess delay with asymmetric robots. Bars and markers are means; error bars are 95% confidence intervals. Values are computed from the released per-episode results.

Noisy bids. Multiplying each bid by a log-normal factor $e^{\sigma\epsilon}$, $\epsilon \sim \mathcal{N}(0,1)$, raises `cost_bid`'s excess delay to 0.2 ± 0.1, 1.5 ± 0.6, 3.3 ± 0.8, and 4.5 ± 0.9 s at $\sigma = 0.1, 0.25, 0.5, 1$ (Fig. 2a). At $\sigma = 0.5$ bidding is no better than the noise-free closest-refuge rule (3.2 ± 0.8 s on the same draws); it is clearly ahead only for $\sigma \le 0.25$.

Asymmetric robots. With differing urgency (1 or 3), payload, or half speed (5 040 episodes), the urgency-weighted excess delay is 27.1 ± 2.6 s for fixed priority, 3.5 ± 0.4 s for the default rulebook, and 0.04 ± 0.02 s for `cost_bid`, which picked the weighted reference's yielder in 98% of episodes (Fig. 2b). The reference that minimizes unweighted delay has 7.0 ± 1.5 s weighted excess delay: the choice of objective matters as much as the protocol.

B. Incentives and fairness

Misreporting under lower-bid-yields. If B inflates its bid by $2\times$ or more, it never yields: its own delay falls from 9.1 ± 1.4 to 8.1 ± 1.2 s while total delay rises by 4.9 s (+20%). Lower-bid-yields bidding is not incentive compatible, and an earlier payment rule we implemented, in which the proceeder paid the yielder's reported bid, was also gameable (A gained 7.5 s by reporting $0.25\times$ its cost, B 0.7 s at $0.75\times$).

Clarke-pivot payments. We scaled one or both robots' reports by $k \in \{0.25, \ldots, 4\}$ (10 values) over 3 bay positions, B urgency 1 and 3, and 60 draws (10 080 episodes, all successful). No unilateral misreport by A or B increased the deviator's utility, its delay plus payment, under the predicted costs; under the realized simulator delays the largest gain was 0.02 s. This is dominant-strategy truthfulness with respect to the shared cost model only. The mechanism is not budget balanced, consistent with [28], [29]: the bank collects 9.3 s (B urgency 1) and 10.5 s (urgency 3) of credit per encounter, which a fleet would have to recycle as a virtual currency. Joint shading is profitable: if both robots report $0.5\times$, their combined utility improves by 4.9 s. Payments are charged in credit, so a robot that expects a large payment may prefer not to participate; we do not analyze individual rationality.

Fairness. Fixed priority makes B yield in every episode. Over 50 sequences of 20 repeated encounters, cost-optimal choice already reaches a Jain index of 0.93 on cumulative delays; fixed priority costs 27% more delay for 0.75, strict alternation 22% for 0.99, and credit-weighted choice, which minimizes total delay plus λ times the difference in cumulative

TABLE I
PROTOCOL ABLATION UNDER 90–99% LOSS AND 2–8 S LATENCY (360 EPISODES PER ROW; THE BAY CHECK HAD NO EFFECT AND IS OMITTED). NO ROW HAD A COLLISION.

Policy	Commit	State	Success [%]	Deadl.	Delay [s]
Fixed priority	✓	✓	99.2	3	89.8 ± 3.9
	✓	–	99.2	3	90.0 ± 4.0
	–	✓	98.1	7	86.0 ± 3.6
	–	–	98.6	5	86.8 ± 3.7
`cost_bid`	✓	✓	98.9	4	79.6 ± 3.9
	✓	–	99.2	3	79.9 ± 3.9
	–	✓	99.4	2	76.4 ± 3.7
	–	–	99.7	1	76.8 ± 3.7

State: state guard. Deadl.: deadlocks. Delay: mean total delay over successful episodes.

delays, reaches 0.997 at $\lambda = 0.25$ for 8% extra delay. This is a simple instance of the credit schemes of [30], [31].

C. Communication impairment

With 6 m detection, loss up to 90%, and latency up to 2 s (3 600 episodes), and with 1.5 m detection and robots at 0.5 m/s (4 320 episodes), every policy succeeded in at least 99.3% of its episodes with no collisions. Under extreme conditions (loss 90–99%, latency 2–8 s, 1 440 episodes) an earlier version of the protocol collided in 6 episodes, each after both robots yielded on inconsistent decisions. On the current code the default protocol has no collisions in these 1 440 episodes and succeeds in 98.9–99.2% per policy.

Table I ablates the three safeguards on the same extreme grid (360 episodes per arm). No arm collided, including the one with all safeguards off (95% upper bound 1.1% per arm), so on the current code we cannot attribute collision avoidance to any single safeguard; the remaining protection comes from the yield and pass logic itself. The lidar bay check never triggered, and its arms were identical episode for episode to those without it. Commit improved fixed priority's success from 98.1–98.6% to 99.2% at 3–4 s extra delay, but did not help `cost_bid`. In 14–24% of episodes both robots started a yield before one gave way.

D. Three robots and junctions

Convoy. With a two-robot convoy against one oncoming robot (72 runs per cell, 0% and 50% loss), a yielding leader could be blocked by its own follower; a "make room" rule, in which the follower backs into its own refuge, removed these deadlocks, and every policy then succeeded in 72/72 runs. With equal urgency, pairwise `cost_bid` ignores the follower and averaged 55.6 s total delay; `group_bid` reduced this to 34.4 s, close to fixed priority and the side-level reference (both 33.7 s). When the lone robot is urgent ($u = 3$), the side-level reference has 85.7 s urgency-weighted delay, fixed priority 90.3 s, `group_bid` 104.7 s, and `cost_bid` 117.1 s: `group_bid` still underestimates what a convoy yield costs.

T-junction. With one robot turning into the branch, one passing, and one leaving the branch (48 runs per cell), naive waiting in the lane deadlocked fixed priority in 19 of 48 runs at 0% loss. Letting the turning robot wait in a free bay before the junction gave 48/48 for fixed priority, the rulebook, and `cost_bid` at 0% and 50% loss, with mean delay over successes of 60.5, 53.1, and 48.9 s at 0% loss. The reactive baseline succeeded in 28/48. All failures in this study were deadlocks.

E. Inconsistent and wrong maps

When robots know different bays (two-bay corridor, 10 seeds per cell), map disagreement caused no collision: commit detects the mismatch and both adopt the lower-ID robot's decision. It costs time when that robot has the poorer map, +9.1 s for `cost_bid` and +29.6 s for the rulebooks; listing known bays in each BID removed the cost in every case tested. When the useful bay was physically blocked, every negotiating policy initially deadlocked (10/10), because the yielder's bay check mistook the obstacle for a robot. Distinguishing obstructions from robots via STATE messages and falling back to the next refuge restored 10/10 success with no collisions, at about +46 s because the obstruction is discovered only on arrival. These samples are small.

F. Learned negotiator

On 150 held-out encounters in the training layout, the learned table reached rulebook-level weighted excess delay (+2.6 s, against +2.5 s for the rulebook, +9.2 s for fixed priority and 0.0 s for `cost_bid`). In an unseen single-bay layout it was no better than fixed priority (+13.1 s against +12.3 s), with 15% of observations unseen, so it does not generalize across layouts.

G. Gazebo validation

Table II summarizes 158 Gazebo runs with one run per scenario, policy, and seed. These runs predate the unified code version (they include the T-junction and occlusion fixes but not the bid-cost correction), so they test transfer of the protocol, not the final numbers. In the single-bay corridor all 40 negotiating runs chose the same first yielder as 2D, with makespans within 0.4 s; the reactive baseline agreed in 5 of 10 runs, as expected from its random back-off, and collided once. Convoy and 50%-loss runs agreed with 2D on the first yielder in every case. The T-junction initially failed in Gazebo for two reasons that the 2D simulator hid: a goal on the room wall was "reached" in 2D, which has no wall perception, and the branch walls occlude the lidar of a robot inside it, so it left the branch in front of an approaching robot. After correcting the goal and letting the exit check also count peers whose STATE shows them approaching, all 36 T-junction runs succeeded with the same first yielder as 2D. Because these fixes were checked on the same cases that revealed them, they are not an independent validation. With `cost_bid` and the robot leaving the branch, Gazebo finished about 22 s faster than 2D despite identical first decisions; preliminary trace analysis points to a renegotiation stall in the 2D runs rather than to Gazebo.

TABLE II
GAZEBO RUNS (FIXED PRIORITY, RULEBOOK, `COST_BID`; PLUS REACTIVE AND A PROXIMITY-FIRST RULEBOOK IN THE SINGLE-BAY SWEEP).

Scenario	Runs	Success	Coll.	Same yielder
Single bay, negotiating	40	40	0	40
Single bay, reactive	10	8	1	5
Convoy (2 v 1)	18	18	0	18
Single bay, 50% loss	18	18	0	18
T-junction, before fixes	36	10	3	30
T-junction, after fixes	36	36	0	36

Coll.: collisions. Same yielder: same (first) yielder as the 2D simulator. The 50% loss runs also have 0.5 s latency.

V. DISCUSSION AND LIMITATIONS

Three conclusions hold across our studies. First, communication is cheap insurance: any explicit agreement removes the deadlocks and back-off "dances" of the reactive baseline. Second, the right rule depends on what the robots know: cost bidding is best when robots differ and their costs are known to within $\sigma \approx 0.25$ in log terms, but it needs payments to resist unilateral misreporting and side-level costs to be efficient for convoys, and even our group bid trails fixed priority when the lone robot is urgent. Third, under extreme packet loss our protocol had no collisions, but the ablation shows that this cannot be credited to the commit step or the bay check alone.

The study has clear limits. All results are from simulation; the Gazebo runs are one per case and predate the final code, and physical TurtleBot3 trials are future work. `cost_bid` and the Clarke mechanism use the simulator's own motion model, so their efficiency and truthfulness hold relative to that model only, and the mechanism neither balances its budget nor resists collusion. The best forced yielder is a reference for the yield decision, not a global optimum, and only the convoy has a three-robot reference. The T-junction is approximated as a deep bay, and the map studies use 10 seeds per cell. The learned baseline is tabular and centrally trained, not deep MARL, and we have not yet evaluated LLM negotiators, a natural next agent family for this benchmark [16], [17].

VI. CONCLUSION

We presented a benchmark for head-on corridor encounters with pull-in bays and compared negotiation policies under identical physics and communication models. Cost bidding with a good cost model matches the best forced yielder and handles asymmetric robots; Clarke-pivot payments remove the incentive for unilateral misreporting at the price of a budget surplus; a small fairness credit makes repeated encounters fair for 8% extra delay; and first-come reservation and fixed priority leave substantial delay on the table. The simulator, ROS 2 nodes, scenarios, study scripts, and per-episode results are released to support comparison with learned and language-model negotiators.

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